

Notes: Bolton and Dewatripont (QJE, 1994)

The Firm as a Communication Network

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1 Big picture

Question. How should a firm organize information processing and communication when agents have limited attention, communication takes time, and specialization makes agents faster at repeated tasks?

Setting. The paper studies an organization that receives a continuing flow of information. Each date brings a new *cohort* of information items. The organization extracts value only after all items in a cohort are processed and aggregated to at least one decision maker.

Core friction. Communication is not free. It takes time for an agent to read, absorb, or listen to information produced by someone else. Processing information is also time-consuming, and no single agent can process the entire incoming flow without becoming overloaded.

Main force. Repetition creates returns to specialization. If an agent repeatedly processes the same type of item, his processing time falls. The benefit of adding more agents is therefore greater specialization; the cost is more communication.

Central trade-off. Bigger organizations allow finer specialization, but they require more messages, more aggregation, and more coordination. Efficient internal structure balances these two forces.

Main theoretical contribution. The paper derives organizational form from a simple communication-cost problem. It shows why efficient networks tend to be centralized and pyramidal even without incentive problems, authority problems, or agency costs.

Main results.

- Efficient networks economize on duplicate communication.
- Each agent should generally report to at most one superior.
- Delegation occurs only when an agent is overloaded.
- Higher-level agents handle more aggregated reports and therefore have fewer direct subordinates.
- Efficient networks may resemble a regular managerial pyramid or a conveyor-belt production line, depending on whether specialization or communication-cost savings dominate.

Economic takeaway. The paper gives a theory of the internal organization of firms based on limited attention and information-processing costs. Hierarchies are not only incentive-control devices; they can also be efficient communication networks.

2 Model environment and key objects

2.1 Organization and objective

The firm is an infinitely lived organization that processes valuable information. It can hire agents from a large pool of identical workers at a given opportunity cost.

The firm's objective is to maximize the flow return from processed information. Since the value of each processed cohort is fixed, the problem becomes minimizing the labor time spent processing and communicating each cohort.

Interpretation. The organization is a production technology for decisions. The "output" is processed information, and the main input is agents' time.

2.2 Cohorts and information items

At every instant, a new information cohort arrives. Each cohort contains M information items and has value $R > 0$ once fully processed.

All items in a cohort must be processed to extract the value R . Cohorts are informationally independent: learning about one cohort does not help process another cohort.

Interpretation. A cohort can be read as a file, a project, a customer order, a production batch, or a decision problem. The M items are the facts, subtasks, parts, or signals that must be handled before the organization can act.

2.3 Processing technology

If processing time is constant, processing one item takes time $\tau > 0$. A single agent processing an entire cohort alone needs τM units of time.

The maximum flow return of one isolated agent is therefore

$$\frac{R}{\tau M}. \quad (1)$$

Interpretation. This is the no-communication benchmark. It avoids coordination costs but sacrifices the gains from dividing labor.

2.4 Communication technology

If agent i sends m_i processed items to agent j , the communication time is

$$C(m_i) = \tau(\lambda + am_i), \quad (2)$$

where

- $\lambda > 0$ is the fixed time cost of establishing or reading a message;
- $a > 0$ is the marginal reading time per item;
- $\lambda + a < 1$, so reading a single already-processed item is faster than processing it from scratch.

Interpretation. Communication has increasing returns in message size. A condensed report with many items has one fixed setup cost, so bundling information can reduce average communication time per item.

2.5 Delay-minimization benchmark

If two agents split a cohort equally, each processes $M/2$ items. One agent then sends his processed items to the other. The time until the receiving agent has the whole cohort is

$$\tau \left(\frac{M}{2} + \lambda + a \frac{M}{2} \right), \quad (3)$$

which can be less than τM .

Interpretation. Parallel processing can reduce delay, but it creates communication. The paper discusses this benchmark but puts more emphasis on throughput and specialization.

2.6 Returns to specialization

With specialization, each cohort is an M -tuple

$$(n_1, n_2, \dots, n_M), \quad (4)$$

where item n_k is a specific type of information.

If an agent processes the same type of item with frequency x , his unit processing time is

$$\tau(x), \quad \tau'(x) < 0, \quad \tau''(x) \geq 0. \quad (5)$$

The same frequency effect applies to reading reports: if an agent repeatedly reads a report containing m_i specific items at frequency x , reading time is

$$\tau(x)(\lambda + am_i). \quad (6)$$

Interpretation. Agents learn by repetition. The more often they handle a task or a report, the faster they become. This is the source of the division of labor inside the firm.

2.7 Information overload

The incoming flow is large enough that no small set of agents can process everything without specialization and delegation. The organization must therefore design a fixed communication network to process cohorts repeatedly.

Interpretation. The firm is needed because the information environment creates overload. The internal network is a way to allocate scarce attention.

2.8 Network terminology

The paper uses the following layer structure.

Layer 0. Agents who only process raw items are in layer 0.

Higher layers. An agent is in layer h if he receives at least one message from an agent in layer $h - 1$ and no message from agents in higher layers.

Pyramidal network. A network is pyramidal if every agent in every layer $h = 0, 1, \dots, H - 1$ sends his information to only one other agent.

Interpretation. The layer notation converts a communication network into a hierarchy. The hierarchy is not based on authority; it is based on the direction of information aggregation.

2.9 Two canonical organizational forms

Regular pyramidal network. A regular pyramid has agents in layer $h > 0$ who only read and aggregate messages. They receive messages from layer $h - 1$, send to one superior in layer $h + 1$, and agents in the same layer have the same number of subordinates.

Conveyor belt. A conveyor belt has only one agent in each layer. Each agent processes the same number of raw items and passes an increasingly aggregated report to the next agent.

Interpretation. The regular pyramid resembles administration and management. The conveyor belt resembles sequential production or assembly-line organization.

3 Model and theoretical results (all equations + interpretation)

3.1 Single-agent specialization benchmark

If one agent processes all M items alone and the frequency of processing is x_1 , then

$$x_1 = \frac{1}{\tau(x_1)M}. \quad (7)$$

Interpretation. The agent processes a new cohort every $\tau(x_1)M$ units of time. Because frequency itself affects processing time, x_1 is determined as a fixed point.

3.2 Two-agent delegation example

Suppose two agents jointly process each cohort. Agent 1 processes m_1 items and sends them to agent 2. Agent 2 processes the remaining $M - m_1$ items and reads agent 1's report.

Agent 1's workload per cohort is

$$\tau(x_2)m_1. \quad (8)$$

Agent 2's workload per cohort is

$$\tau(x_2)(M - m_1 + \lambda + am_1). \quad (9)$$

If workloads are equalized, then

$$\tau(x_2)m_1 = \tau(x_2)(M - m_1 + \lambda + am_1). \quad (10)$$

Solving gives

$$m_1^* = \frac{M + \lambda}{2 - a}. \quad (11)$$

The corresponding processing frequency satisfies

$$x_2 = \frac{1}{\tau(x_2)m_1^*}. \quad (12)$$

Interpretation. Equal workloads make both agents fully employed. Delegation raises frequency and specialization, but it also forces the receiver to spend time reading a report.

3.3 Why delegation is limited

If agent 1 processes more than m_1^* , agent 1 becomes the bottleneck and agent 2 has idle time. This is inefficient because the same communication link is used while the potential specialization gain is not fully exploited.

If agent 1 processes less than m_1^* , agent 2 becomes the bottleneck. This may save communication because agent 2 directly processes more items, but it lowers the frequency of repeated tasks.

Interpretation. Delegation should not be maximized mechanically. The efficient amount of delegation depends on the shape of $\tau(x)$ and on communication costs.

3.4 General workload notation

For agent i , define

- y_i : number of raw items processed by i ;
- n_i : number of reports read by i ;
- M_i : total information content of the reports read by i .

At frequency x , agent i 's workload per cohort is

$$\tau(x)(y_i + n_i\lambda + aM_i). \quad (13)$$

After normalizing $\tau(x) = 1$ for the communication-network comparison, define the maximum workload as

$$T^* = \max_i \{y_i + n_i\lambda + aM_i\}. \quad (14)$$

Interpretation. The network's feasible frequency is limited by the most loaded agent. Organizational design is therefore about distributing processing and reading loads while avoiding redundant messages.

3.5 Proposition 1: agents should not have too much slack

Result. In any efficient network:

1. an agent in layer $h \geq 2$ has workload at least $T^* - \lambda$;
2. an agent in layer $h \geq 1$ has workload at least $T^* - 1 + a$.

Interpretation. If a higher-layer agent has enough unused capacity, the network can reroute a message directly to him and save one communication transit. If a nonbottom agent has enough slack, the network can move some raw processing to him and save communication. Efficient networks therefore do not leave high-layer agents with large amounts of unused attention.

3.6 Proposition 2: efficient networks are pyramidal

The paper imposes a mild indivisibility condition, Property (U): information that enters a report cannot be costlessly disaggregated and sent around the reporting agent without changing the task structure.

Result. All nonpyramidal networks satisfying Property (U) are dominated by some pyramidal network. They are strictly dominated when an agent with several direct superiors is in any layer higher than 1.

Interpretation. Multiple direct superiors duplicate communication. Splitting the agent's reporting tasks across separate agents preserves information flow while reducing unnecessary routing. This gives a communication-cost foundation for the familiar rule that each worker generally reports to one boss.

3.7 Proposition 3: no redundant communication links

Consider n agents $(1, \dots, n)$ who all communicate directly with the same superior.

Result. In any efficient network, it must be infeasible to divide these n agents' total workload among only $n - 1$ agents without exceeding the workload constraint T^* .

Interpretation. If one subordinate can be eliminated without overloading the remaining subordinates, the superior can read one fewer report and the organization saves the fixed cost λ . Efficient networks therefore avoid unnecessary direct reports.

3.8 Proposition 4: higher layers handle larger reports

Let $m_i(h)$ be the smallest report received by an agent i in layer h . Let $M_j(h - L)$ be the largest report received by one of i 's direct or indirect subordinates j in layer $h - L$.

Result. In any efficient pyramidal network,

$$m_i(h) > M_j(h - L), \quad L = 1, \dots, h - 1. \quad (15)$$

Interpretation. The smallest report handled by a higher-layer agent should still be larger than the largest report handled by lower-layer subordinates. If not, reports can be swapped to reduce communication time. This formalizes the idea that higher management deals with more aggregated information.

3.9 Proposition 5: limited skip-level reporting

Suppose all layer-0 agents have the same workload y .

Result. In any efficient network, layer-0 agents communicate with superiors in layer 1 or layer 2.

Interpretation. A bottom-layer processor should not normally skip many layers. Extreme skip-level reporting sends small raw reports directly to very high levels, violating the principle that upper layers should handle more aggregated reports.

3.10 Regular pyramids

A regular pyramid is useful when it is efficient for bottom-layer agents to specialize fully in processing and for higher-layer agents to specialize fully in aggregation.

In the special case with $a = 0$, let

$$M = n^H = \left(\frac{1}{\lambda}\right)^H, \quad (16)$$

where n is the number of subordinates per agent in the regular pyramid and H is the number of layers.

The regular pyramid with full specialization processes cohorts at frequency x^* satisfying

$$x^* = \frac{1}{\tau(x^*)}. \quad (17)$$

Interpretation. Layer-0 agents each process one item, and higher-layer agents aggregate reports. The parameter restriction makes it possible for all agents to have full workload in this special benchmark.

3.11 Conveyor belt

In the conveyor belt, agents form a chain. Each agent processes one item and adds it to the accumulated report. The frequency z^* satisfies

$$\frac{1}{z^*} = \tau(z^*)(1 + \lambda). \quad (18)$$

Interpretation. Relative to the regular pyramid, the conveyor belt uses fewer communication links. But most agents both process and aggregate, so they may specialize less sharply.

3.12 Proposition 6: comparing the conveyor belt and the regular pyramid

Under full specialization and $a = 0$, total labor time under the regular pyramid is

$$L_{\text{pyr}} = \tau(x^*) \left[M + \lambda \left(M + n^{H-1} + n^{H-2} + \cdots + n \right) \right]. \quad (19)$$

Total labor time under the conveyor belt is

$$L_{\text{conv}} = \tau(z^*) [M + \lambda(M - 1)]. \quad (20)$$

Result. The conveyor belt is more efficient than the regular pyramid if and only if

$$\tau(z^*) [M + \lambda(M - 1)] < \tau(x^*) \left[M + \lambda \left(M + n^{H-1} + \cdots + n \right) \right]. \quad (21)$$

Interpretation. The conveyor belt economizes on the fixed costs of communication links. The regular pyramid may be better if it preserves enough specialization. Organizational form is chosen by comparing communication-cost savings against specialization losses.

3.13 Positive marginal reading costs and the limits of the conveyor belt

When $a > 0$, the conveyor belt has an additional drawback: early items pass through many agents before reaching the top. The network may fail to minimize the average number of transits.

Interpretation. With positive marginal reading costs, not only the number of messages matters; the number of times each item is reread also matters. This pushes efficient networks toward pyramidal aggregation rather than a pure chain.

3.14 Proposition 7: exact regular pyramids are rare

Result. Regular pyramidal networks with more than three layers in which all agents have a full workload do not exist whenever $a > 0$.

Interpretation. Integer constraints make perfectly regular pyramids hard to sustain. Real organizations may look approximately pyramidal, but exact regularity is not generally efficient once marginal reading costs are positive.

3.15 Proposition 8: when a full-workload regular pyramid exists

Result. Consider regular pyramidal networks in which agents in layer 0 each process a single item and all agents have a full workload. Such networks strictly dominate all other networks with the same processing frequency.

Interpretation. When exact full-workload regularity is feasible, a regular pyramid is powerful because it combines full specialization with no redundant communication. The caveat is that the feasibility conditions are restrictive.

4 What makes the argument credible (assumptions, interpretation, and limits)

4.1 Why hierarchy emerges without incentives

The model does not rely on moral hazard, monitoring, authority, or ownership. Hierarchy emerges because information must be aggregated, messages are costly, and duplicate communication wastes attention.

Economic message. The paper isolates a non-incentive reason for firms to be hierarchical: a hierarchy can be the least costly way to aggregate dispersed information.

4.2 Why centralization saves communication

If several agents all need the same information, the organization may duplicate messages many times. Centralizing information in one place can reduce repeated transmission.

Economic message. Centralization is efficient when the cost of transmitting information is high relative to the cost of letting one agent read a more complete report.

4.3 Why delegation occurs only under overload

An agent delegates because his own attention is scarce. If he is not close to overloaded, it is cheaper for him to process or absorb more information directly than to create an additional communication link.

Economic message. Delegation is a response to limited attention, not an automatic virtue. More delegation can increase specialization, but it can also increase communication costs.

4.4 Why higher layers have fewer subordinates

Higher-layer reports contain more aggregated information. Reading each report therefore takes more time. To keep workloads balanced, higher-layer agents must have smaller spans of control.

Economic message. The model predicts that the span of control should shrink near the top of the hierarchy. Top managers handle broader reports, so they cannot efficiently have as many direct reports as lower-level supervisors.

4.5 Why information technology can flatten firms

Lower communication costs reduce both the fixed cost λ and the marginal reading cost a . When communication becomes cheaper, one manager can absorb more information without adding layers.

Economic message. Better communication technology can lead to flatter organizations. This need not mean decentralization; it can also mean that senior managers can centralize more information because communication is cheaper.

4.6 What the model abstracts from

The model deliberately leaves out many features of real organizations:

- incentive conflicts and monitoring;
- private benefits of control;
- endogenous authority and decision rights;
- changing environments and adaptive reorganization;
- multidimensional expertise and team complementarities;
- strategic distortion of messages.

Interpretation. These omissions are not mistakes; they clarify the information-processing mechanism. But they also mean the model should be read as a theory of one important force, not a complete theory of firm boundaries or managerial control.

4.7 Relation to the management literature

The results connect to classic management questions about span of control, hierarchy depth, assembly lines, and the effect of communication technology. The model provides formal logic for several familiar observations:

- organizations are often pyramidal;
- senior managers tend to have fewer direct reports;
- production lines can be efficient when sequential aggregation saves communication links;
- communication technology can reshape internal hierarchy.

5 Figures and key theoretical results

5.1 Figure I: duplication and delay

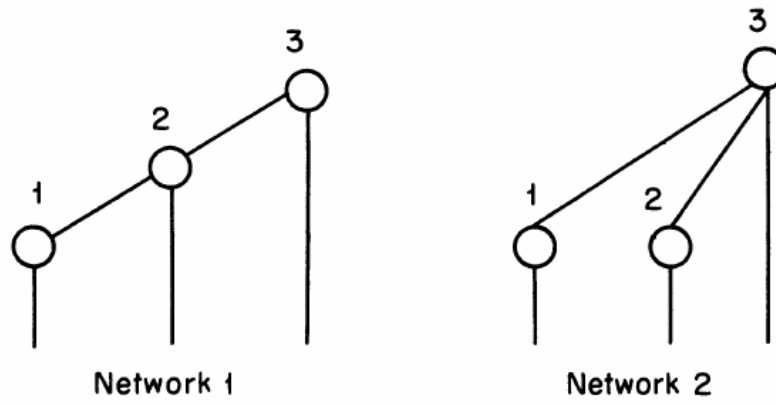


FIGURE I

Figure 1: Two communication networks with three agents.

Read:

- Network 1 sends information through agent 2 before it reaches agent 3.
- Network 2 sends both agents' information directly to agent 3.
- Network 1 creates duplication: information from agent 1 is communicated twice.
- Network 2 reduces delay because it avoids unnecessary transits.

Economic message. Even simple networks differ in communication efficiency. Avoiding redundant transits is a basic reason for centralizing information aggregation.

5.2 Figure II: returns to specialization

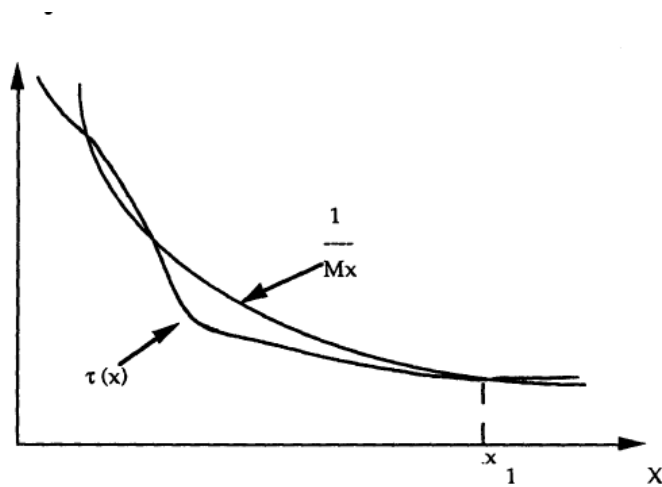


FIGURE II

Figure 2: The fixed point for processing frequency under returns to specialization.

Read:

- The downward-sloping curve captures lower unit processing time at higher frequency.
- The fixed point determines the feasible rate at which cohorts can be processed.
- Higher frequency improves specialization but may require more delegation and communication.

Economic message. Specialization is endogenous. The organization chooses a network partly to raise the repetition frequency of tasks.

5.3 Figure III: the conveyor belt

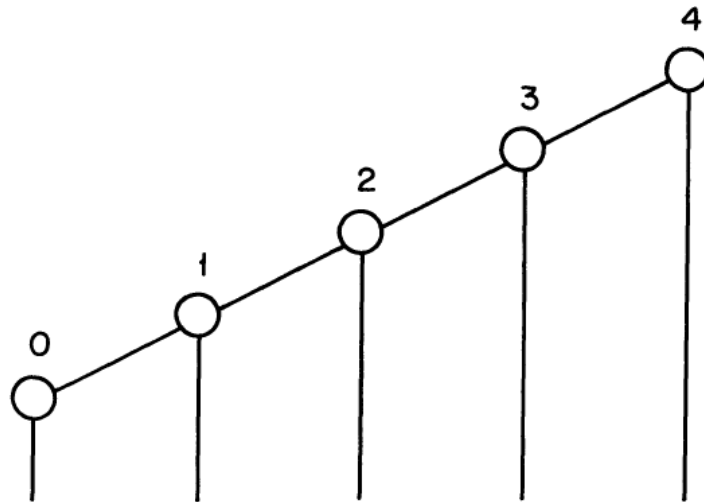


FIGURE III

Figure 3: A conveyor-belt network.

Read:

- There is one agent in each layer.
- Each agent adds processed information to the accumulated report.
- The chain minimizes the number of direct communication links.
- It may be efficient when communication-link costs are high and marginal transit costs are not too large.

Economic message. The same framework can explain both administrative hierarchies and assembly-line-like production structures.

5.4 Summary of main propositions

Result	Formal content	Economic meaning
Proposition 1	Higher-layer agents cannot have too much slack.	Efficient networks reroute or reassign work to save communication.
Proposition 2	Nonpyramidal networks are dominated under Property (U).	One-agent-one-superior reporting avoids duplicate communication.
Proposition 3	Direct subordinates cannot be consolidated without exceeding T^* .	No redundant communication links.
Proposition 4	Higher-layer agents receive larger reports than their subordinates.	Information becomes more aggregated as it moves upward.
Proposition 5	Equal-load bottom agents report only to layer 1 or 2.	Skip-level reporting is limited.
Proposition 6	Conveyor belt beats regular pyramid iff its total labor time is lower.	Choose form by comparing specialization losses and communication savings.
Proposition 7	Exact full-workload regular pyramids with more than three layers do not exist when $a > 0$.	Perfect regularity is fragile under marginal reading costs.
Proposition 8	If a full-workload regular pyramid exists, it dominates same-frequency alternatives.	Regular pyramids are efficient when feasibility conditions align.

Table 1: Main theoretical results and their economic interpretation.

6 Short summary

- The paper models the firm as a network for processing and communicating information.
- Information arrives continuously in cohorts of M items, and all items must be processed to create value.
- Communication takes time: each message has a fixed cost λ and a marginal reading cost a per item.
- Specialization makes agents faster when they repeatedly process the same type of item.
- The central trade-off is specialization versus communication.
- Efficient networks avoid duplicate communication and tend to be centralized.
- Agents generally report to one superior, so efficient networks are pyramidal.
- Delegation is efficient only when it relieves overload.
- Higher layers handle more aggregated reports and have fewer subordinates.
- Depending on parameters, efficient networks may look like managerial pyramids or conveyor belts.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Bolton and Dewatripont show that hierarchical organization can arise from communication and information-processing costs alone. If agents learn by specializing but must communicate processed

information, efficient organizations centralize information flow, limit redundant reporting, and often take pyramidal form.

7.2 Economic intuition

Specialization raises productivity, but specialization fragments information. Communication reunifies that information, but communication consumes attention. The optimal organization is the network that gets the specialization benefits while minimizing unnecessary messages and transits.

7.3 Takeaway

The paper is a foundational theory of firms as communication networks. Its main lesson is that hierarchy, span of control, delegation, and even assembly-line organization can be understood as responses to limited attention and costly information aggregation.